Dickson Odhiambo Owuor

Curriculum Vitæ

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Personal Profile

Dickson is a computer scientist whose research centers on **data-centric knowledge discovery**. His primary contributions lie in **gradual pattern mining**, where he has advanced and optimized mining algorithms and extended their applicability to **time-series** analysis and **Symbolic Aggregate Approximation (SAX)**-based models.

Through active collaboration with internationally recognized researchers, Dickson has broadened his work to include *graph theory based approaches for microscopy image analysis, ontology-driven knowledge representation using knowledge graphs, deep learning for text-based disease surveillance,* and *reinforcement learning for bioinformatics*. He is also the maintainer of several open-source research libraries, including the *so4gp* Python library (`https://pypi.org/project/so4gp/`) and the *sgtlib* Python library (`https://pypi.org/project/sgtlib/`).

At **Strathmore University**, Dickson is a **Senior Lecturer** with over 10 years of experience teaching undergraduate-level and graduate-level Computer Science courses. He has co-supervised **3 PhD candidates to completion**, supervised **10+ MSc theses** and **50+ undergraduate projects**, and is currently co-supervising three PhD candidates. In addition, he serves as the Vertically Integrated Project **(VIP) Program Coordinator** and plays a key role in establishing and strengthening local and international industry–academic partnerships within the School of Computing and Engineering Sciences (SCES).

Research Interests

Areas	Artificial Intelligence · Machine Learning · Reinforcement Learning · Text Mining · Swarm Intelligence · Knowledge Representation · Graph Theory · Meta-Heuristics · Gradual Pattern Mining
Thematic Areas	Bioinformatics · Disease Surveillance · Feature Selection · Microscopy Image Analysis · Time-series Analysis

Education

2018–2020 **Ph.D. in Computer Science**, *Montpellier University*, in Montpellier

- Thesis Title: *Capturing temporal constraints of gradual patterns.*
- Thesis Advisors: Anne Laurent (LIRMM, Université de Montpellier) and Joseph Orero (SCES, Strathmore University).
- Defence Jury: Pascal Poncelet (LIRMM, Univ. Montpellier), Nicolas Sicard (LRIE, l'EFREI), Marie-Jeanne Lesot (LIP6, Sorbonne Université) and Maria Rifqi (LEMMA, Univ. Paris II).

2013–2015 **MSc. in Mobile Telecommunications & Innovation**, *Strathmore University*, in Nairobi, Kenya

* Major Coursework: applied entrepreneurship, mobile & USSD application development, software modeling & system design, mobile & web security, wireless technologies, 2G/3G networks.
* Dissertation Title: *An expert system for screening infectious diseases at airports.*
* Dissertation Advisor: Joseph Onderi Orero (Strathmore University).

2008–2012 **BSc. in Electronics & Computer Engineering**, *Jomo Kenyatta University of Agriculture and Technology*, Juja, Kenya, *First Class Honors*

* Major Coursework: circuit and network theory, analogue electronics, digital electronics, physical electronics, power electronics, quantum electronics, electro-magnetics, control engineering, integrated circuits, microprocessors, signals and communication, numerical and complex analysis, low-level programming, structured programming, compiler construction.

Experience

Research & Teaching Roles

Since 2025 **Senior Lecturer**, *Strathmore University · SCES*, (`https://strathmore.edu/`)

* Actively collaborating in and extending research in the following areas: gradual patterns and their applications, graph-based microscopy image analysis, ontology- and knowledge graph–based knowledge representation, deep learning for disease surveillance, and reinforcement learning for bioinformatics.
* Supervising BSc. projects. Advising MSc., Ph.D. dissertations/theses. Serving as a peer reviewer for international journals and conferences.
* Teaching multiple undergraduate-level and graduate-level courses.

2024–2026 **Visiting Scholar & Academic Affiliate**, *University of Michigan · Kotov Lab (ChemE)*, (`https://kotov.engin.umich.edu/`)

Worked full-time and contributed to research on graph theory based analysis of SEM and confocal microscopy images. Published or assisted in the publication of 3+ high impact papers. Involved in the development of `StructuralGT` and `ResponseGT` software libraries.

2021–2024 **Lecturer**, *SCES · Strathmore University*, Nairobi

* Published or assisted in the publication of 8+ papers. Served as a peer reviewer for several high-ranking journals and conferences.
* Supervised 20+ BSc. projects and, advised 10+ MSc. dissertations/theses to completion. Co-advised 1 Ph.D. thesis to completion. Mentored 5+ students who successfully completed their undergraduate degrees.
* Accumulated 3,000+ hours of teaching practice spanning multiple undergraduate and graduate-level courses, including: *Computer Programming, Embedded Systems and Internet of Things, Distributed Systems, Data Mining, Storage and Retrieval.*

2016–2020 **Doctoral Fellow**, *SCES · Strathmore University*, Nairobi

* Undertook my doctoral research at the Université de Montpellier.
* Assisted in the delivery of undergraduate-level courses, including: *Introduction to Programming, Object-oriented Programming, Embedded Systems and IoT.*
* Supervised 30+ Computer Science-related BSc. projects to completion.

Administrative Roles

Since 2024 **Coordinator**, *VIP Program · Strathmore University*, Nairobi, (`https://sces.strathmore.edu/vip-strathmore/`)
In charge of launching and coordinating the Vertically Integrated Project (VIP) program at Strathmore University.

Since 2021 **Partnerships Lead**, *SCES · Strathmore University*, Nairobi
Responsible for forging research-driven academic and industrial partnerships with local and international institutions. Facilitated academic collaborations between Strathmore University and the following universities: Université de Montpellier (France), University of Michigan (USA), Universidad de Castilla-La Mancha (Spain), and Georgia Institute of Technology (USA).

2015–2020 **Attachment Office Administrator**, *SCES · Strathmore University*, Nairobi
Coordinated undergraduate student internships: service-based learning and work-based learning.

2014–2015 **Graduate Assistant**, *Faculty of IT · Strathmore University*, Nairobi
Assisted in delivery of academic tutorials and in coordinating student internships.

Projects & Awards

Collaborations

Since 2024 **FLOW Summer School**, *Université de Montpellier*
Contributed to the *establishment of a formal academic partnership* between Polytech, Université de Montpellier, and SCES, Strathmore University, resulting in signed Memoranda of Understanding (MoUs). So far we have co-organized **two editions of the FLOW Summer School** (2024–2025) in collaboration with **Prof. Eric Anglaret** (Deputy Director for International Affairs, Polytech Montpellier) and his team. The programme involved approximately **80 undergraduate students** from Strathmore University and supported academic exchange, intercultural training, and student mobility. The collaboration has led to **two Master's admissions** at the Université de Montpellier.

Since 2024 **Kotov Lab**, *Michigan University*
Ongoing collaboration with **Prof. Nicholas Kotov** on a *multidisciplinary research project at the interface of computer science and chemical engineering*. The research focuses on improving the efficiency and accuracy of `StructuralGT`, a graph theory based software for microscopy image analysis, through dynamic search strategies, including swarm intelligence and reinforcement learning. To date, this collaboration has resulted in **three journal publications** (one under review) and the release of `sgtlib`, a Python library published on PyPI.

Since 2024 **Universidad de Castilla–La Mancha (UCLM)**
Ongoing collaboration with **Prof. José Antonio Gámez** and **Prof. Antonio Puerta** on *counter-factual explanations using Bayesian networks*. This work is conducted within the framework of the Ph.D. project of **Kelvin Kibunja**, whom we co-supervise.

Since 2022 **CIRAD, France**
Ongoing collaboration with **Prof. Mathieu Roche**, **Roberto Interdonato**, and **Prof. Sarah Valentin** on *information mining and surveillance systems for food safety and disease monitoring*. Outcomes include the successful Ph.D. supervision and defense of **Edmond Menya** (2025), submission of a **SAAFARI grant proposal**, and **5+ peer-reviewed publications**.

Since 2021 **Université Paris 8**
Ongoing collaboration with **Prof. Nicolas Jouandeau** on the *application of reinforcement learning algorithms to bioinformatics and financial data analysis*. This collaboration has led to the successful Ph.D. supervision and defense of **Stephen Obonyo** (2024) and one ongoing Ph.D. project (**Daniel Ruiru**).

Since 2019 **Siemens AG, Germany**

Ongoing collaboration with **Prof. Thomas Runkler** on the *optimization of gradual pattern mining algorithms using swarm intelligence techniques*. This collaboration has resulted in **3+ peer-reviewed publications**.

Since 2018 **LIRMM, France**

Long-term collaboration with **Prof. Anne Laurent** across data mining, knowledge representation, and data lake architectures. Outcomes include a **CNRS-funded project**, the successful Ph.D. supervision and defense of **Deperias Kerre** (2025), and **10+ peer-reviewed publications**.

Grants Received

Since 2021 **French Government Ph.D. Sponsorship**, *French Embassy in Kenya*

Played a key role in securing French Government funding, through the Embassy of France in Kenya, to **support doctoral studies at universities in France**. The sponsorship scheme requires a principal supervisor based in France and a local co-supervisor at Strathmore University. I served as the local co-supervisor and coordination lead for the following awarded sponsorships:

- Paris 8 University (2024): *in collaboration with N. Jouandeau, secured funding for Daniel Ruiru.*
- University of Montpellier (2022): *in collaboration with A. Laurent, secured funding for Deperias Kerre.*
- University of Montpellier (2022): *in collaboration with M. Roche, secured funding for Edmond Menya.*
- Paris 8 University (2021): *in collaboration with N. Jouandeau, secured funding for Stephen Obonyo.*

2021 **DSCA Afrique**, *CNRS*

Dispositif de Soutien à la Coopération avec l'Afrique sub-saharienne

- Amount: *20,000 euros.*
- Description: *awarded to support an existing research collaboration between Strathmore University (Kenya) and Montpellier University (France) in the form of individual exchanges between researchers working in SCES (Kenya) and LIRMM (France), research workshops, conference and publication fees.*

2019 & 2020 **Student Travel Grant**, *EUSFLAT*

Supported participation in the LFA 2019 Conference (Alès, France) and the LFA 2020 Conference (Sète, France).

2018 **French Government Doctoral Scholarship**, *French Embassy in Kenya*

Awarded to pursue a Ph.D. in Computer Science at the Université de Montpellier (2018–2020).

2013 **Safaricom Master's Scholarship**, *Strathmore University*

Awarded to pursue a Master of Science degree (2013–2015).

Awards Received

2011 **Vice Chancellor's Academic Award**, *JKUAT*

Awarded for outstanding academic excellence as the top student.

Committees

IPMU **International Conference on Information Processing and Management of Uncertainty in Knowledge-Based Systems**, *2026*

Reviewer – *Participated in the paper review process.*

CARI **African Conference on Research in Comp. Science & Applied Mathematics**, *2024-26*

Program Committee (PC) Member

- *Participates in the paper review process, ensuring the selection of high-quality research contributions.*
- *Contributes to the evaluation and decision-making process for conference submissions.*

KDIR **International Conference on Knowledge Discovery and Information Retrieval**, *2021*
Reviewer – *Participated in the paper review process.*

International Journal Reviews

`JMLC` · International Journal of Machine Learning and Cybernetics · *Springer*

`Sci Rep` · Scientific Reports · *Springer Nature*

`KIS` · Knowledge and Information Systems · *Springer*

`TETA` · Journal of Experimental & Theoretical Artificial Intelligence · *Taylor & Francis*

`IS` · Information Systems · *ScienceDirect*

`JSC` · The Journal of Supercomputing · *Springer Nature*

Memberships

IEEE *Institute of Electrical and Electronics Engineers* · `8+ Year Membership` *No: 95329588*

EBK *Engineers Board of Kenya* · `2+ Year Membership` *No: B27868*

Ph.D. Supervisions

Completed

1. `(Defense Date - Nov 2025)` Deperias Kerre · *Mining the Hetero-Structural Design and the Optoelectronic Characteristics of Quantum Cascade Lasers* · 33% co-supervision · Co-supervisors at Université de Montpellier: Anne Laurent and Kenneth Maussang · `2022-2025 sponsorship by French Embassy in Kenya.`

2. `(Defense Date - Jul 2025)` Edmond Menya · *Enhancing Thematic Features for Event-based Surveillance Systems* · 33% co-supervision · Co-supervisors at Université de Montpellier: Mathieu Roche and Roberto Interdonato · `2022-2025 sponsorship by French Embassy in Kenya.`

3. `(Defense Date - Dec 2024)` Stephen Obonyo · *Bioinformatic Computation With Active Learning* · 50% co-supervision · Co-supervisor at Université de Paris 8: Nicolas Jouandeau · `2021-2024 sponsorship by French Embassy in Kenya.`

Ongoing

1. `(Tentative - Dec 2027)` Daniel Ruiru · *Resource Allocation Problems with Deep Reinforcement Learning* · 50% co-supervision · Co-supervisor at Université de Paris 8: Nicolas Jouandeau · `2024-2027 sponsorship by French Embassy in Kenya.`

2. `(Tentative - Dec 2028)` Kelvin Kibunja · *Counterfactual explanations using Bayesian Networks* · 33% co-supervision · Co-supervisors at Universidad de Castilla-La Mancha: Gámez Martín and Puerta Callejón.

Technical Skills

ML & AI Developing and optimizing ML models (with expertise in NLP, RL, DRL) ●●●●○

DL Models Experience with CNNs, RNNs, DNNs, Transformers ●●●◐○

Frameworks PyTorch ●●●●◐

Languages	C/C++, Python, MatLab, R, Java	●●●●●
Markdown	HTML, CSS, TailwindCSS	●●●●●
Scripting	JavaScript, NodeJS, ReactJS, Angular	●●●◐○
Typesetting	LaTex	●●●●●
Ontology	OWL	●●●◐○

Languages

English	Proficient	●●●●●
Swahili	Native	●●●◐○
French	Basic	●○○○○

Hobbies

Basket ball · Swimming · Running · Meditation · Blogging · Reading

Publications

Bibliometric Indicators

Last updated on September 09, 2026.

Indicator	Scopus	Google Scholar
Publications	27	38
Citations	77	129
h-index	5	7
i10-index	–	3

Articles in Peer-Reviewed Journals

[1] Edmond Menya, Roberto Interdonato, Dickson Owuor, and Mathieu Roche. Novelty detection in event surveillance documents. *IEEE Access* *(Scimago Ranking: Q1)*, pages 1–1, 2026. doi:10.1109/ACCESS.2026.3666022

[2] Connor N. McGlothin, Kody G. Whisnant, Emine Sumeyra Turali Emre, Dickson Owuor, Jun Lu, Xiongye Xiao, Drew Vecchio, Scott Van Epps, Paul Bogdan, and Nicholas Kotov. Autocatalytic nucleation and self-assembly of inorganic nanoparticles into complex biosimilar networks. *Angewandte Chemie International Edition* *(Scimago Ranking: Q1)*, 64(9):e202413444, 2025. doi:10.1002/anie.202413444

[3] Deperias Kerre, Anne Laurent, Kenneth Maussang, and Dickson Owuor. A concise ontological model of the design and optoelectronic properties in the quantum cascade laser domain. *Semantic Web* *(Scimago Ranking: Q1)*, 16(4):22104968251359870, 2025. doi:10.1177/22104968251359870

[4] Edmond Menya, Roberto Interdonato, Dickson Owuor, and Mathieu Roche. Explainable epidemiological thematic features for event based disease surveillance. *Expert Systems with Applications* *(Scimago Ranking: Q1)*, 250:123894, 2024. doi:10.1016/j.eswa.2024.123894

[5] Stephen Obonyo, Nicolas Jouandeau, and Dickson Owuor. Self-Playing RNA Inverse Folding. *SN Computer Science (Scimago Ranking: Q2)*, 5(4):403, 2024. doi:10.1007/s42979-024-02659-x

[6] Dickson Odhiambo Owuor, Thomas Runkler, Anne Laurent, and Lesley Bonyo. Clustering-based gradual pattern mining. *International Journal of Machine Learning and Cybernetics (Scimago Ranking: Q1)*, 2023. doi:10.1007/s13042-023-02027-w

[7] Dickson Odhiambo Owuor, Thomas Runkler, and Anne Laurent. A metaheuristic approach for mining gradual patterns. *Swarm and Evolutionary Computation (Scimago Ranking: Q1)*, pages 101–205, 2022. doi:10.1016/j.swevo.2022.101205

[8] Dickson Odhiambo Owuor, Anne Laurent, and Joseph Onderi Orero. Mining fuzzy temporal gradual emerging patterns. *International Journal of Uncertainty, Fuzziness and Knowledge-Based Systems (Scimago Ranking: Q3)*, pages 655–676, 2021. doi:10.1142/S0218488521500288

[9] Dickson Odhiambo Owuor, Thomas Runkler, Anne Laurent, Joseph Orero, and Edmond Menya. Ant colony optimization for mining gradual patterns. *International Journal of Machine Learning and Cybernetics (Scimago Ranking: Q1)*, pages 1–21, 2021. doi:10.1007/s13042-021-01390-w

International Conference Proceedings

[1] Dickson Odhiambo Owuor and Anne Laurent. Enhanced Mining of FTGPs Using Mutual Information and Metaheuristic Technique. In *2025 IEEE International Conference on Fuzzy Systems (FUZZ)*, pages 1–6, 2025. doi:10.1109/FUZZ62266.2025.11197674

[2] Deperias Kerre, Anne Laurent, Kenneth Maussang, and Dickson Odhiambo Owuor. Semantic enrichment of the quantum cascade laser properties in text-a knowledge graph generation approach. In *LLM-TEXT2KG 2025 - 4th International Workshop on LLM-Integrated Knowledge Graph Generation*, volume 4020 of *CEUR Workshop Proceedings*, pages 68 – 89, 2025. URL: https://hal.science/lirmm-05245564

[3] Edmond Menya, Mathieu Roche, Roberto Interdonato, and Dickson Owuor. EpidGPT: A Combined Strategy to Discriminate Between Redundant and New Information for Epidemiological Surveillance Systems. In Amon Rapp, Luigi Di Caro, Farid Meziane, and Vijayan Sugumaran, editors, *Natural Language Processing and Information Systems*, pages 439–454, Cham, 2024. Springer Nature Switzerland. doi:10.1007/978-3-031-70239-6_30

[4] Sarah Valentin, Edmond Menya, Roberto Interdonato, Mathieu Roche, and Dickson Owuor. Integrating textual data for enhanced explanation of food crises at subnational scale. In *Discovery Science - Late Breaking Contributions 2024*. CEUR Workshop Proceedings, 2024

[5] Deperias Kerre, Anne Laurent, Kenneth Maussang, and Dickson Owuor. A text mining pipeline for mining the quantum cascade laser properties. In Alberto Abelló, Panos Vassiliadis, Oscar Romero, Robert Wrembel, Francesca Bugiotti, Johann Gamper, Genoveva Vargas Solar, and Ester Zumpano, editors, *New Trends in Database and Information Systems*, pages 393–406. Springer Nature Switzerland, 2023

[6] Edmond Menya, Mathieu Roche, Roberto Interdonato, and Dickson Owuor. Enriching epidemiological thematic features for disease surveillance corpora classification. In *Proceedings of the 13th Language Resources and Evaluation Conference*, pages 3741–3750. European Language Resources Association, 2022

[7] Dickson Odhiambo Owuor and Anne Laurent. Efficiently mining large gradual patterns using chunked storage layout. In *Advances in Databases and Information Systems*, pages 30–42, Cham, 2021. Springer International Publishing. doi:10.1007/978-3-030-82472-3_4

[8] Dickson Odhiambo Owuor, Anne Laurent, and Joseph Onderi Orero. Exploiting IoT data crossings for gradual pattern mining through parallel processing. In *ADBIS, TPDL and EDA 2020 Common Workshops and Doctoral Consortium*, pages 110–121, Cham, 2020. Springer International Publishing. doi:10.1007/978-3-030-55814-7_9

[9] Dickson Owuor, Anne Laurent, and Joseph Orero. Mining fuzzy-temporal gradual patterns. In *2019 IEEE International Conference on Fuzzy Systems (FUZZ-IEEE)*, pages 1–6, New York, NY, USA, june 2019. IEEE. doi:10.1109/FUZZ-IEEE.2019.8858883

National Conference Proceedings

[1] Dickson Owuor, Anne Laurent, Joseph Orero, and Olivier Lobry. Gradual pattern mining tool on cloud. In *Extraction et Gestion des Connaissances: Actes EGC'2021*. BoD-Books on Demand, January 2021

[2] Dickson Owuor, Anne Laurent, and Joseph Orero. Croisements de données de l'internet des objets pour l'extraction de motifs graduels flous. In *29 Rencontres francophones sur la logique floue et ses applications (LFA 2020)*, pages 29–38. Cépaduès Éditions, november 2020

[3] Dickson Owuor, Anne Laurent, and Joseph Orero. Extraction de motifs graduels temporels flous. In *28 Rencontres francophones sur la logique floue et ses applications (LFA 2019)*, pages 1–12. Cepadues Editions, november 2019

Software Publications

[1] Dickson Owuor (Ph.D.). owuordickson/structural-gt: v3.6.8, Oct 2025. doi:10.5281/zenodo.17417897

[2] Dickson Owuor (Ph.D.). owuordickson/sogp_pypi: v0.7.1, Aug 2025. doi:10.5281/zenodo.16764372

Dataset Publications

[1] Deperias Kerre, Anne Laurent, Kenneth Maussang, and Dickson Owuor. An instruction dataset for extracting quantum cascade laser properties from scientific text. *Data in Brief*, 58:111255, 2025. doi:10.1016/j.dib.2024.111255

[2] Edmond Menya, Roberto Interdonato, Dickson Owuor, and Mathieu Roche. Novelty dataset (animal health, food security, climate change), 2025. doi:10.18167/DVN1/8C9ZTN

[3] Edmond Menya, Roberto Interdonato, Dickson Owuor, and Mathieu Roche. PADI-web corpus used for the EpidBioELECTRA approach, 2023. doi:10.18167/DVN1/WD1UC2